Core Finance Week 3

MGMP 543

Prof. Benedict Guttman-Kenney

Spring 2025

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Jones Graduate School of Business

Welcome Back! Today's Topics

- Week 2 Recap
  - Examples
- Equities
  - What is Equity?
  - Valuing Equities
  - Capitalization Rates
  - Growth Opportunities

Week 3 Introductions

Week 3 Introductions

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Week 3 Introductions

- Your Name
- Favorite food? Favorite restaurant in Houston / Texas / US / the world?

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Week 2 Recap

What Did We Learn Last Week?

What Did We Learn Last Week? Covered A LOT!

Capital Budgeting

- Capital budgeting is process to evaluate major projects/investments
- Net Present Value is very similar to Week 1 Present Value
- Capital budgeting methods

  (Net Present Value, Internal Rate of Return & Hurdle Rates, Payback Period, ROI, Profitability Index, Equivalent Annual Cost)

Bonds

- Pricing Zero-Coupon and Coupon Bonds. Premium/Par/Discount. Very similar to week 1 annuities.
- Yield to Maturity (YTM) is IRR if you hold to maturity
- US Treasuries, Yield Curve, Inverted Yield Curves, Bond Defaults

Quiz Question: An investor is considering investing in a new business. The business is expected to require a cash investment of $1.5 million today. The business will generate yearly (after-tax) cash flows of $1 million for four years, starting one year from today. The CEO has calculated that the NPV of the business is $1.57 million. What is the required rate of return as a percent to one decimal place?

Recap on Multiple IRRs

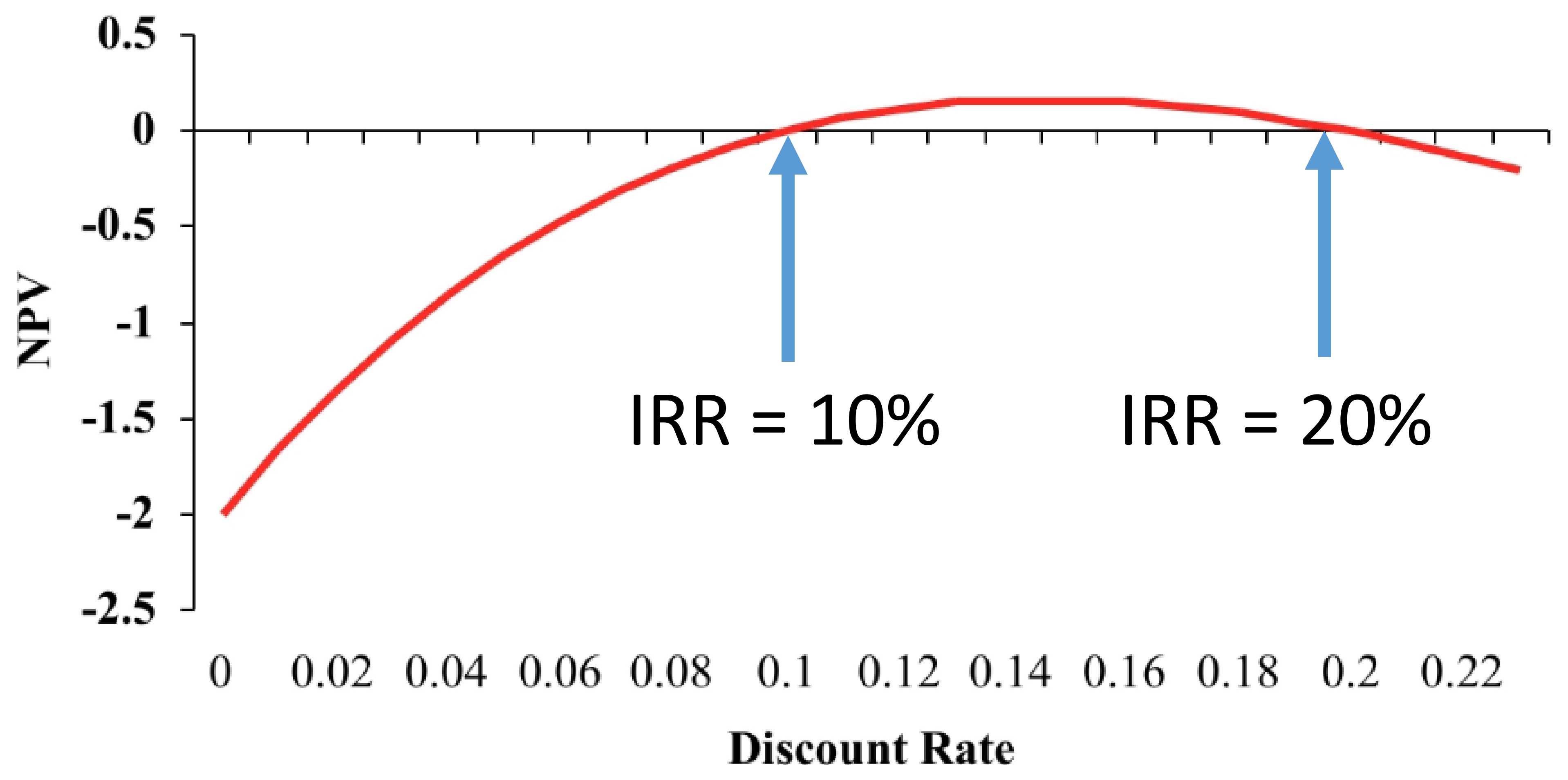
Internal Rate of Return (IRR) is the interest rate that sets the NPV = 0.

Hurdle Rate (HR) is the organization's required rate of return on an investment.

Invest if IRR > HR

- If HR is 7%, and IRR is 10% or 20%. Invest!
- If HR is 15% and IRR is 10% don't invest. But if IRR is 20% invest!
- If HR is 25% and IRR is 10% or 20%. Don't invest.

Project	CF_0	CF_1	CF_2	NPV (r = 10%)	IRR
A	-100	230	-132	0	10% or 20%



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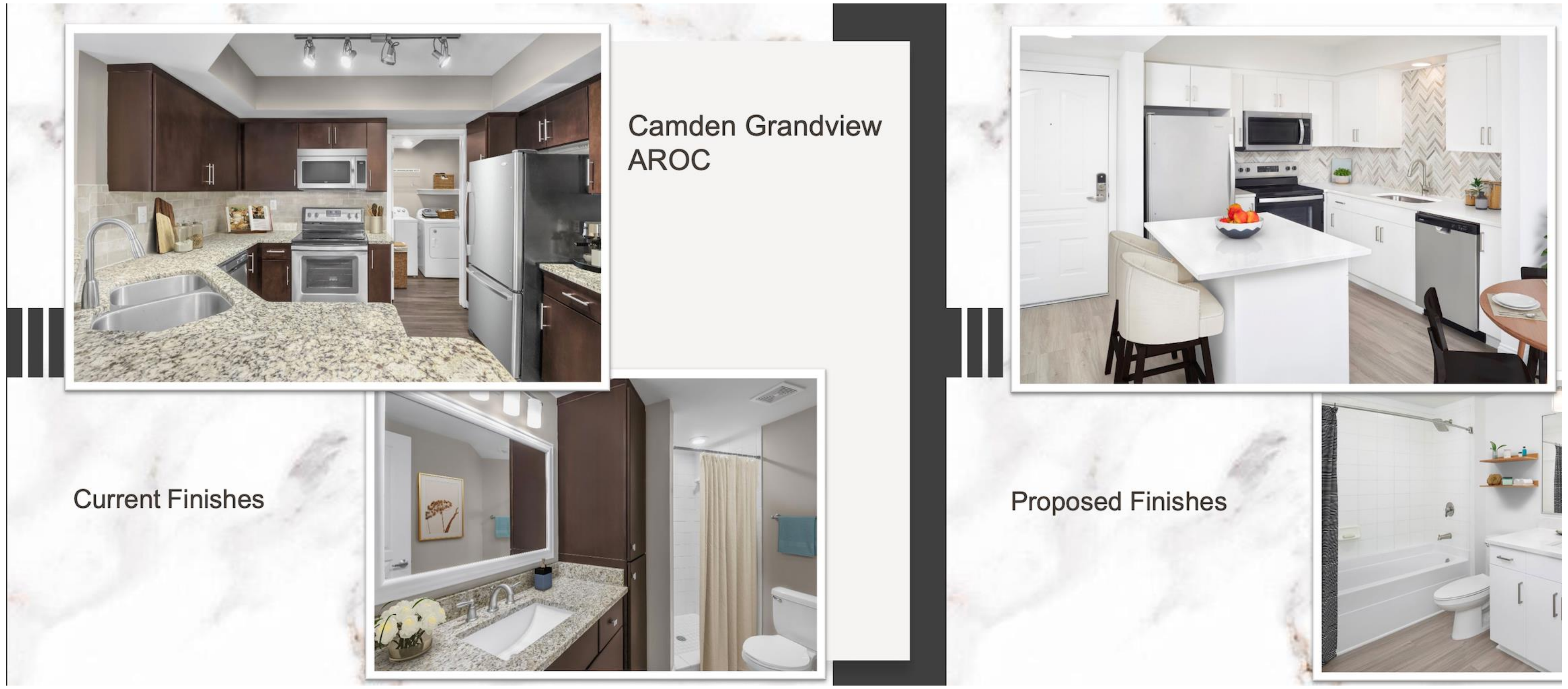
IRR Example from MBA Classmate: Real Estate Upgrade



Request: $50,260,149 for unit repositions at 6 communities.

Total Units to Reposition: 1,709

Total Projected Cost of Repositions: $50,260,149

Total Projected Cost per Unit: $29,409

Average Monthly Premium: $173

Projected ROIC: 7.1%

Projected Unlevered IRR: 10.4%

[1] *FFO Impact is calculated using total rental income from reposition premiums and turn cost savings, less vacancy loss due to repositions and total cost of debt. Assumes 5.3% long term cost of debt, per IC parameter's accretion hurdle.*

[5] *The IRR model assumes revenue and expense savings per turn over the 42-month project timeline from unit reposition premiums, turn cost savings, and interior capex savings. The preceding revenue and expense savings is reduced by the reposition costs over the project timeline and by adding 7 days of additional vacancy loss per turn. The revenue from reposition premiums is held flat with no assumed growth and then capped at the end of year-10 by rates from the most recent NAV analysis, per REI.*

3. Pinewood Studios is considering which new studio to invest in. Both studios will generate the same revenues. Which is a better investment at an 8% discount rate?
A) Studio Alec costs $150,000 and is expected to last 4 years with operating costs of $24,000 per year.
B) Studio Boris costs $100,000 and is expected to last 2 years with operating costs of $9,000 per year.

a) Studio Alec
b) Studio Boris
c) Indifferent
d) Not enough information

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b) Studio Boris

Alternative 1: Studio Alec (Trevelyan)
Net Present Value:

$$NPV = -150,000 - \frac{24,000}{0.08}\left[1 - \frac{1}{(1.08)^4}\right] = -229,491$$

Equivalent annual cost (using the annuity formula):

$$-229,491 = \frac{EAC}{0.08}\left[1 - \frac{1}{(1.08)^4}\right]$$

$$EAC = -69,288$$

Alternative 2: Boris (Grishenko)
Net Present Value:

$$NPV = -100,000 - \frac{9,000}{0.08}\left[1 - \frac{1}{(1.08)^2}\right] = -116,049$$

Equivalent annual cost (using the annuity formula):

$$-116,049 = \frac{EAC}{0.08}\left[1 - \frac{1}{(1.08)^2}\right]$$

$$EAC = -65,077$$

Since alternative A is more expensive (i.e. has a higher cost) than alternative B, we should choose Alternative B.

5. What is the price of a bond issued by Oddjob Corporation that pays $100 every year for 4 years, and also pays $1,000 at the end of four years? Assume the discount rate is 12%.

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$$= \frac{C}{r}\left(1 - \frac{1}{(1+r)^T}\right) + \frac{F}{(1+r)^T} = \frac{100}{0.12}\left(1 - \frac{1}{(1+0.12)^4}\right) + \frac{\$1{,}000}{(1+0.12)^4}$$

$939.25

Equities

Equity is Ownership Interest in a Firm

Different to Accounting ‘Shareholder Equity = Total Assets – Total Liabilities’

What is Stock?

What is Stock?

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Jargon-Busting Stocks

- **Stock/Equity** is ownership of a corporation (all the shares of ownership of a corporation). “blue-chip stock” refers to large, well-established company.
- **Share** is a unit of stock ownership.
- **Par Value:** Value of stock when it is issued
- **Book Value:** Value of stock on balance sheet
- **Market Value:** Observed current market price
- **Dividends:** Cash payments regularly made by corporations to stockholders
- **Stock Dividend:** Payment by corporation in stock shares instead of cash
- **Stock Split:** Issuance of larger number of shares in proportion to existing shares outstanding

Types of Stock

Common Stock

- ***Owners*** of the company

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- ***Residual value*** on assets and proceeds of a firm (claim after creditors & preferred stockholders)

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- **Initial Public Offering (IPO)** is first day a private company's stock becomes available to the public to purchase

Preferred Stock

- ***Hybrid*** form of financing, combining features of debt and common stock
- ***Middle priority of*** payment – above common stocks, below bonds
- ***Pays dividends****,* specified and known in advance, usually percentage of par value

  Dividends are cumulative - If firm omits dividend, it must pay it later
- ***No voting rights*** unless company is unable to pay dividends as specified
- Dividends paid from after-tax profits so corporation taxed, and then these dividends are also taxed as income for the preferred stockholders (“double taxation”)

“Convertible Preferred Stock” is preferred stock that can be exchanged for common stock at a pre-agreed fixed conversion ratio.

Who Gets Paid? Equity Payoff Structure

Valuing Equities

Example: Sweetgreen Inc.

Example: Sweetgreen Inc.

IPO Date: 18 November 2021

Valuation at IPO: $2,976,680,000

Money Raised at IPO: $364,000,000

Stock Symbol: $SG

Shares Outstanding: 106,310,000

If you own 1 share in Sweetgreen Inc.'s common stock, you own 0.0000009406% of the SG

Should we buy Sweetgreen Inc?

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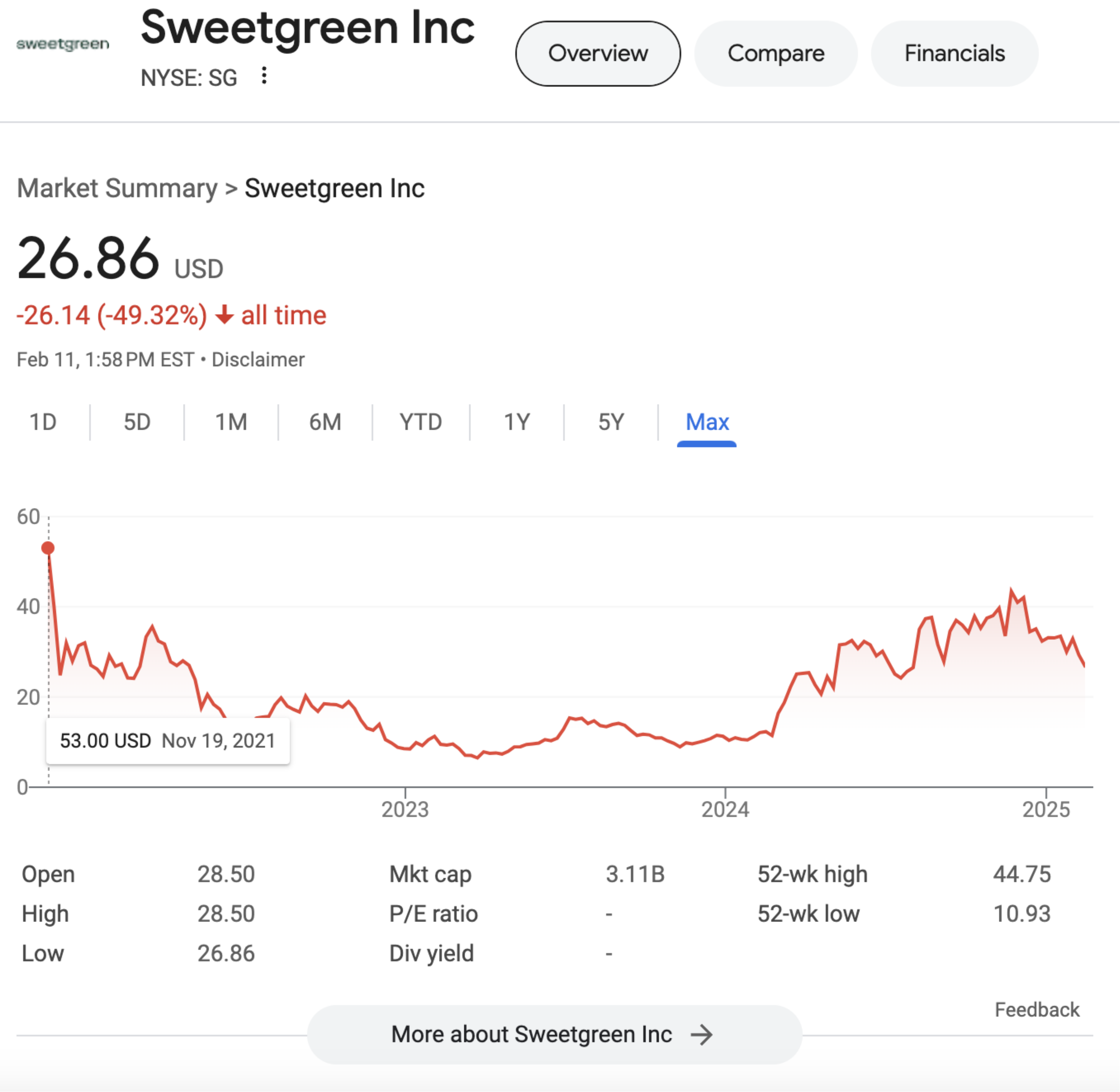


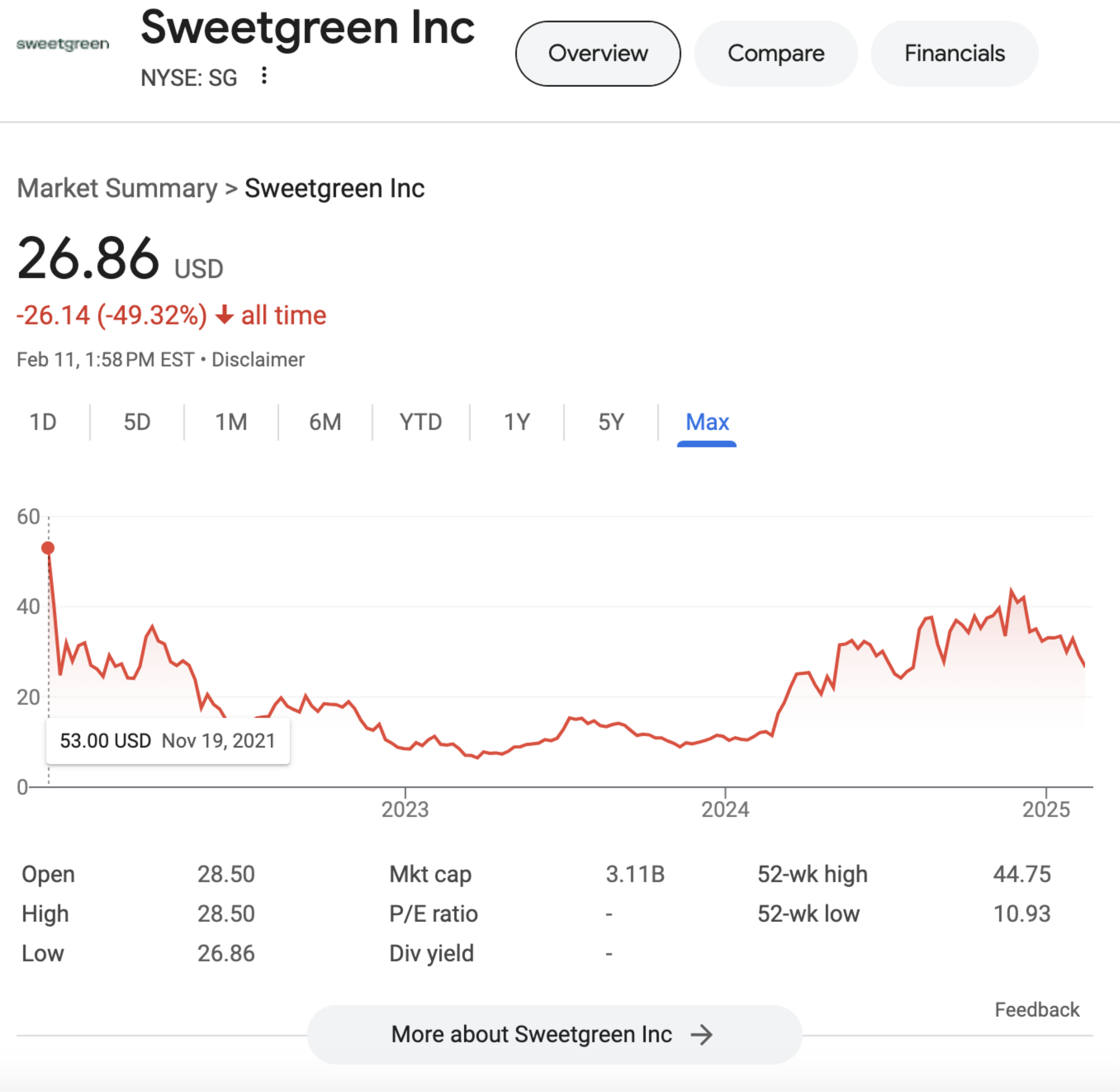
Should we buy Sweetgreen Inc?

Depends on the value of equity

- **Is company worth more than market believes?**

  If so, can buy low and sell high for profit!

- Past price ⬆️ does ***not*** guarantee future price ⬆️



Stock Valuation Notation

- **P:** Stock price
- **r:** Discount Rate
- **D:** Dividend Payment
- **g:** Growth Rate of Dividend Payment
- **EPS:** Earnings Per Share
- **ROE:** Return on Equity
- **y:** Dividend Payout Ratio
- **b:** Plowback Ratio

Price of Equity

- Value of equity is the **discounted expected value of future dividends**. (cashflows expected to be received by the investors from the firm)

- If dividends are growing at a constant rate (g) and the discount rate is greater than the growth rate (r>g) then we can use the **Gordon Growth Formula**:

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$$P_0 = \frac{D_1}{1+r} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

- If dividends are growing at a constant rate (g) and the discount rate is greater than the growth rate (r>g) then we can use the **Gordon Growth Formula**:

Assuming $D_t = D_{t-1}(1+g)$ and r>g

$$P_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

Price of Equity: Example

Assuming $D_t = D_{t-1}(1+g)$ and r>g

$$P_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

Prue Leith Common Stock is currently selling for \$28 per share and has paid dividends of \$2.00 per share over the most recent 12 months.

The consensus expected growth rate of dividends among investment analysts is 3% per year.

If the discount rate is 11%, would you find this stock a good investment?

What Moves Equity Prices?

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

$$P_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

Capitalization Rate

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- If you observe current stock price, and assume price is based on dividend value methodology then we can solve for r.
- Required rate of return is the capitalization rate
- Assuming constant dividend growth we get **capitalization rate**:

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- Popular way to estimate market's required return. For example, used by regulators to set price of providing utilities to infer a reasonable return on capital
- Dividend yield is how much you earn on dividend, per dollar of stock price

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- If you observe current stock price, and assume price is based on dividend value methodology then we can solve for r.
- Required rate of return is the capitalization rate
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$$r = \frac{D_1}{P} + g = \text{dividend yield + growth rate}$$

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- Dividend yield is how much you earn on dividend, per dollar of stock price

Present Value of Growth Opportunities (PVGO)

Estimating Dividend Growth

- CEOs allocate shareholder capital and earn Return on Equity (ROE)
  - ROE: Net income to stockholders divided by book equity
- Hard to predict expected future growth rate of dividends.
- If we assume:

(1) firm does not issue new shares

(2) earns constant ROE, and

(3) retains ("plowback") a constant fraction (b) of its earnings to finance asset growth (& operating income)

Then **dividend growth rate is:**

$$g = ROE \times b$$

Where $b = 1 - \frac{D}{EPS}$

What if b=0? b=1?

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Present Value of Growth Opportunities (PVGO)

- Growth does not create value. Undertaking NPV>0 investments creates value.
- Value only created if retained earnings earn a greater return than the required rate of return
- Price of stock is capitalized value of expected earnings under a non-growth policy **PLUS** the assets in place **PLUS** growth options

$$P = \frac{EPS_1}{r} + PVGO$$

where $PVGO = \frac{b \times EPS_1 \times \left(\frac{ROE}{r} - 1\right)}{r - g}$

Example: If Paul Hollywood Corporation is expected to earn 18% ROE and plowback one-third of its earnings, what would be the sustainable growth on an annual basis?

	Book Value (Start of Year)	Net Income	Dividend	Retained Earnings	Book Value (End of Year)
1	$100				
2					

Example: If Paul Hollywood Corporation is expected to earn 18% ROE and plowback one-third of its earnings, what would be the sustainable growth on an annual basis?

	Book Value (Start of Year)	Net Income	Dividend	Retained Earnings	Book Value (End of Year)
1	**$100**	**18**	**12**	**6**	**106**
2	**$106**	**19.08**	**12.72**	**6.36**	**112.36**

$$g = ROE \times b = 18\% \times \frac{1}{3} = 6\%$$

$$\frac{112.36}{106} - 1 = 6\%$$

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Example: Hammond and Fielding Corporation

- ROE = 0.21 and reinvestment rate (b) is 0.6. What is the growth rate (g)?
- $EPS_1 = \$2.80$, what would next year's dividend be?
- r=14%, what is the stock price using the constant growth rate model?
- What is the price of the stock using PVGO?

Example: Hammond and Fielding Corporation

- ROE = 0.21 and reinvestment rate (b) is 0.6. What is the growth rate (g)?

$$g = ROE \times b = 0.21 \times 0.6 = 12.6\%$$

- $EPS_1 = \$2.80$, what would next year's dividend be?

$$D_1 = 2.80 \times (1 - 0.6) = \$1.12$$

- r=14%, what is the stock price using the constant growth rate model?

$$D_0 = \frac{D_1}{r - g} = \frac{\$1.12}{0.14 - 0.126} = \$80$$

- What is the price of the stock using PVGO?

$$P = \frac{2.80}{0.14} + \frac{0.6\,(2.80)\,(\frac{0.21}{0.14} - 1)}{0.14 - 0.126} = \$20 + \$60 = \$80$$

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Amazon Barnes & Nobel Over 20 Years

	Amazon		Barnes & Nobel	
	1997	2017	1997	2017
Share Price	$18.5	$968	$50.75	$7.6
Shares Outstanding	23.9 m	480 m	33.8 m	72.6 m
Market Value	$442 m	$464,640 m	$1,715 m	$551 m
Book Value (Total Shareholder Equity)	$45.6 m	$23,214 m	$468 m	$555.8 m
Market-to-Book Value	9.7	20.02	3.67	0.99

Why pay 20 market-to-book value to own Amazon? Why only 1 for Barnes & Nobel?

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Valuations So Far Based on Dividends

- What about the price of stocks that do not pay dividends?
  - Alphabet paid no dividends until 2024. P = $0?

Valuations So Far Based on Dividends

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  - Alphabet paid no dividends until 2024. P = $0?

- Instead need to work out the cash flow stream of expected future dividends.

- **Week 9 Preview:** Adjust formulas to value Free Cash Flows (FCF) instead of dividends.

Stock Market

Stock Exchanges

- **Stock Exchange.** A place where you can exchange your stock.
    - US: New York Stock Exchange, Nasdaq
    - UK: London Stock Exchange
    - China: Shanghai Stock Exchange
    - Japan: Japen Exchange Group (Tokyo Stock Exchange)
    - Europe: Euronext, German Stock Exchange (Deutsche Borse AG)
    - And many others but 21 exchanges with $1+ trillion market capitalization, trade most world's equities.
- **Primary** Market where issue shares (IPO), **Secondary** Market where trade shares.
- **Dealers** (like auto dealers) hold inventory

    - with **bid** price willing to buy, **ask** price willing to sell. **Spread** = bid – ask.
- **Brokers** (like real estate brokers) don't hold inventory but bring buyers & sellers together.

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Stock Market Index

A Stock Market Index measures performance of a series of stocks.

- Dow Jones Industrial Average - 30 stocks
- Standard & Poor's 500 (S&P500) – 500 companies' stocks (503 stocks)
- Nasdaq Composite Index – 2,500 stocks
- Wilshire 5000, ~3,300 stocks

Large varieties of indexes for different sectors / geographies
e.g., S&P500 Energy Sector Index.

Usually, indexes are weighted by market capitalization (i.e., larger firms get more weight), S&P500 and Nasdaq, and Wilshire. Whereas Dow Jones is price-weighted (i.e., higher priced stocks get more weight), and you can also see equal-weighted versions (i.e., each stock gets equal weight).

It's 1802. What do you think has the highest return over the next 220 years?

a) Bonds
b) Cash
c) Gold
d) Stocks

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It's 1802. What do you think the average annual real return on stocks is over the next 220 years?

a) <0%
b) ~0%
c) 1-2%
d) 3-4%
e) 5-6%
f) 7-8%
g) 9-10%
h) >10%

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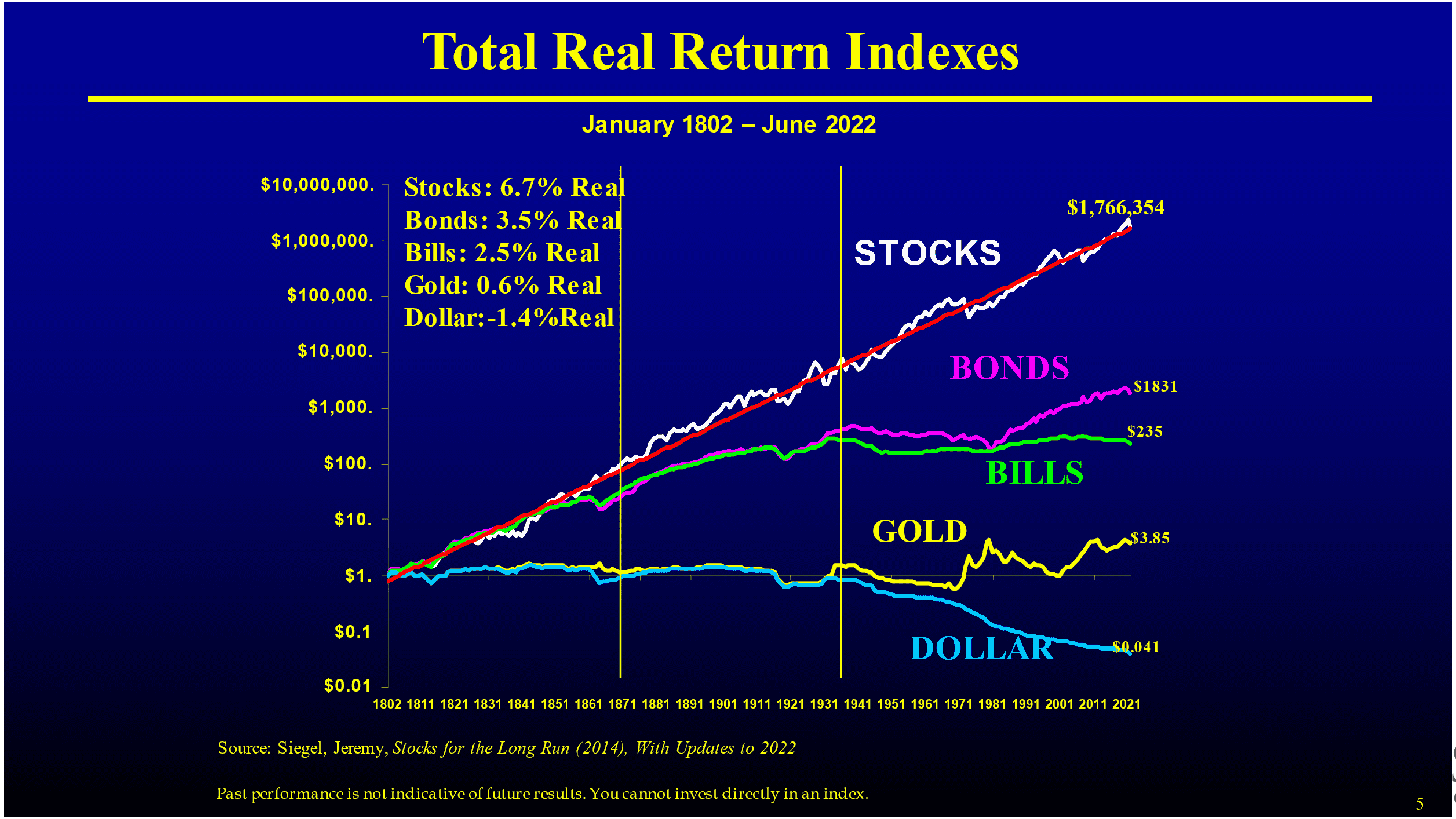
How Do Stocks Perform Historically?



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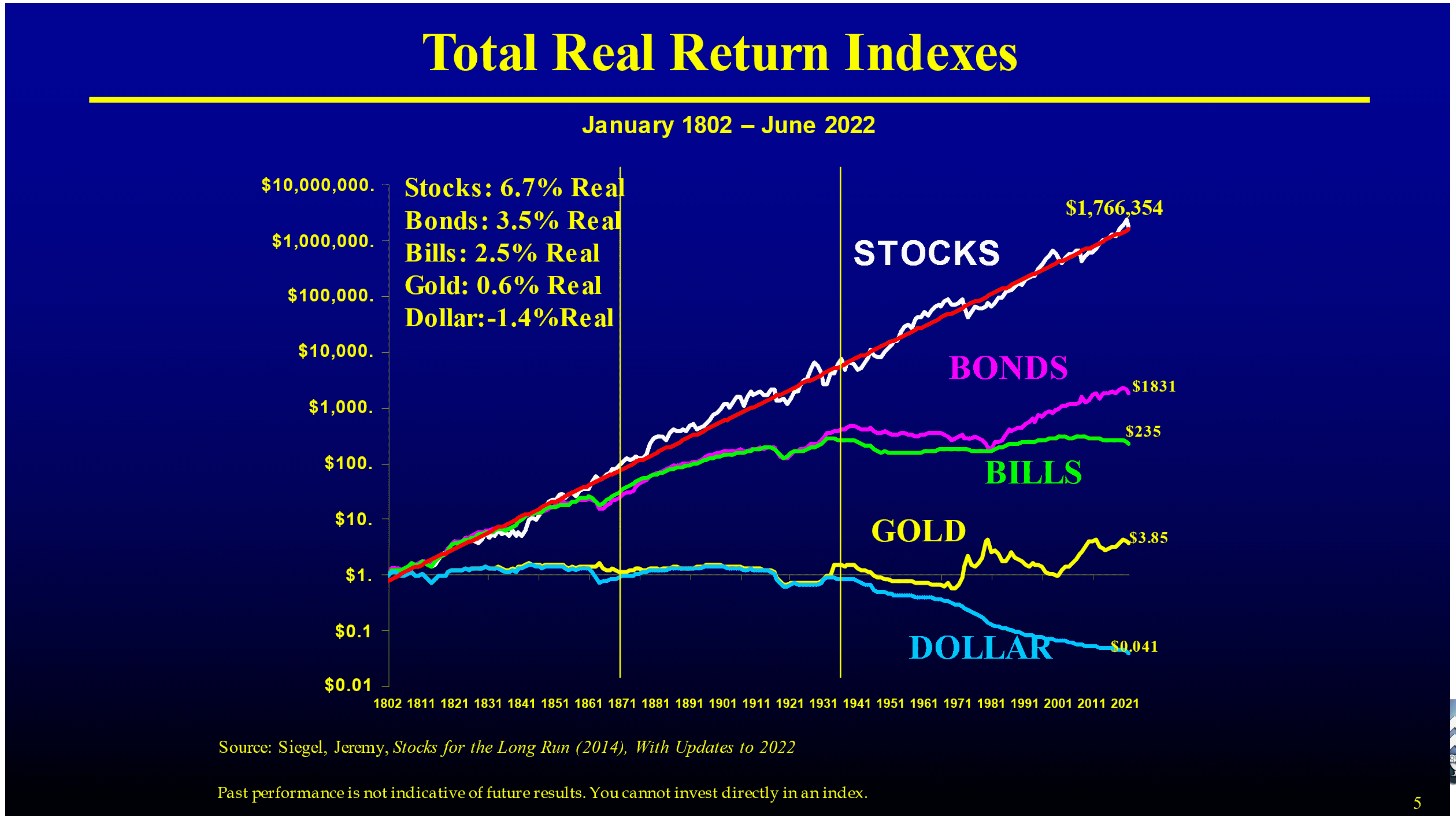
How Do Stocks Perform Historically?

- Stocks outperform bonds, bills, gold, and cash.
- Long-run real return on stocks is ~ 6.7%



The "Sage" of Omaha

"The one thing I will tell you is the worst investment you can have is cash. Everybody is talking about cash being king and all that sort of thing.

Cash is going to become worth less over time.

But good businesses are going to become worth more over time.

And you don't want to pay too much for them so you have to have some discipline about what you pay. But the thing to do is find a good business and stick with it. We always keep enough cash around so I feel very comfortable and don't worry about sleeping at night. But it's not because I like cash as an investment. **Cash is a bad investment over time. But you always want to have enough so that nobody else can determine your future essentially."**

How Did Stocks Perform Recently?
S&P500 Last Friday 28 February 2025

Stocks are traded continuously

but here are prices each minute

'Candlestick' charts
(five minute intervals)

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How Did Stocks Perform Recently?
S&P500 **Last Week** to Friday 28 February 2025

Stocks are traded continuously

but here are prices each minute

'Candlestick' charts
(15 minute intervals)

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How Did Stocks Perform Recently?
S&P500 **Last Month** to Friday 28 February 2025

Stocks are traded continuously

but here are prices each minute

‘Candlestick’ charts
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How Do Stocks Perform Recently? S&P500

Last 6 months to 28 February 2025

Last 12 months to 28 February 2025

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How Do Stocks Perform Recently?
S&P500 since 1995

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What Do We Learn?

What Do We Learn?

- Over long horizons, stock market index outperform bonds, cash, gold.
- But stock markets have larger fluctuations in prices.
- Returns vary depending on when stock bought and sold (market timing) and time horizon for holding.

Next week…

- What about risks and returns on individual stocks vs. stock market index?

Individual Stock Example

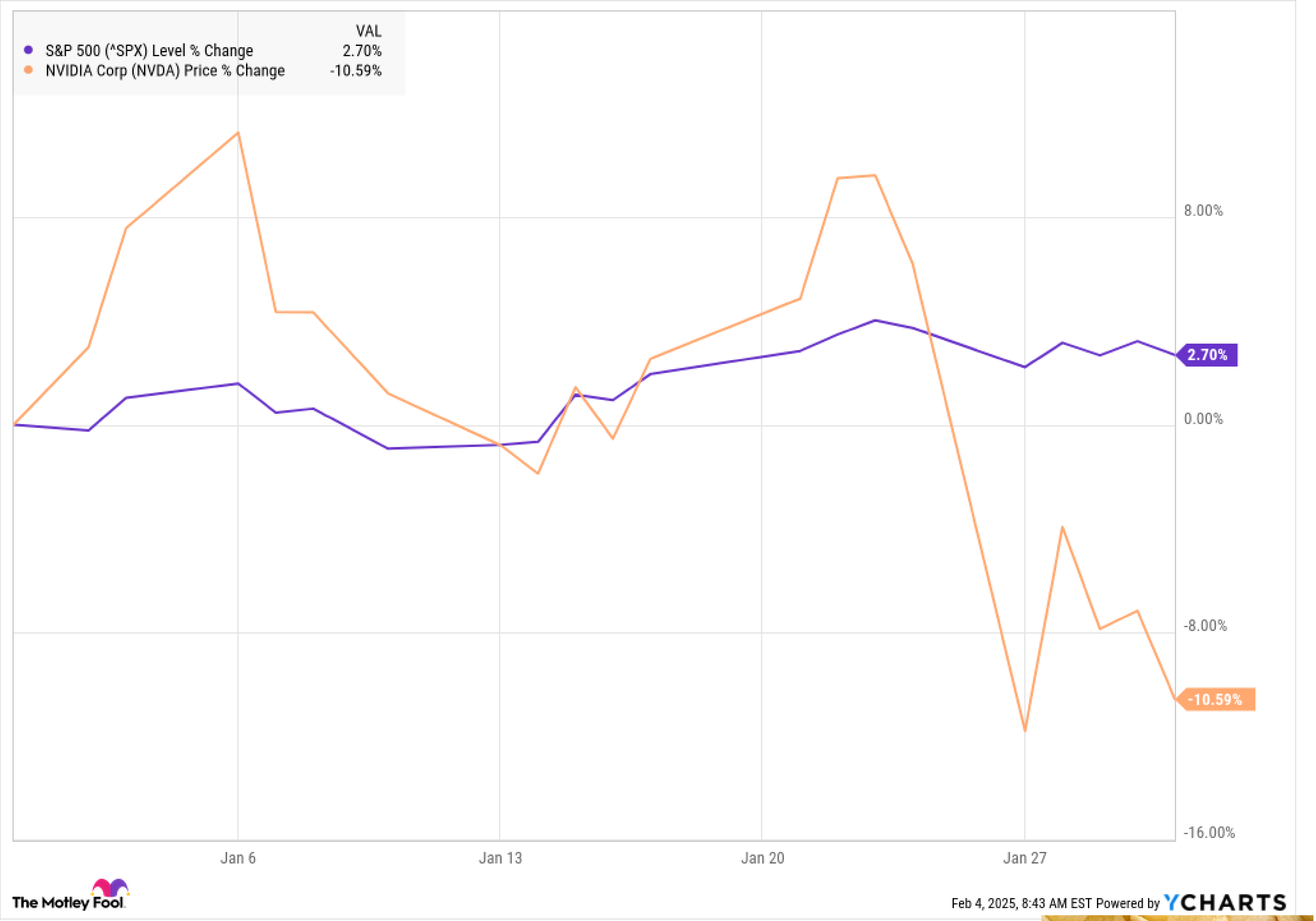
NVIDIA Corp (NASDAQ:NVDA)

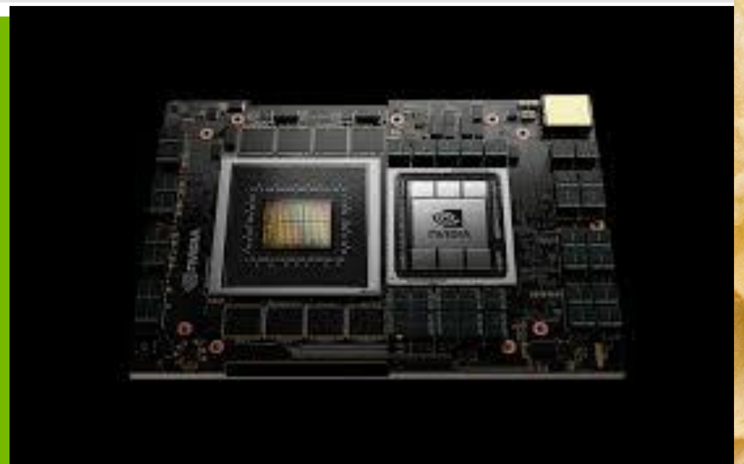
Monday 27th January 2025

Stock price dropped 17%.

⬇$600 bn market capitalization!

Why?





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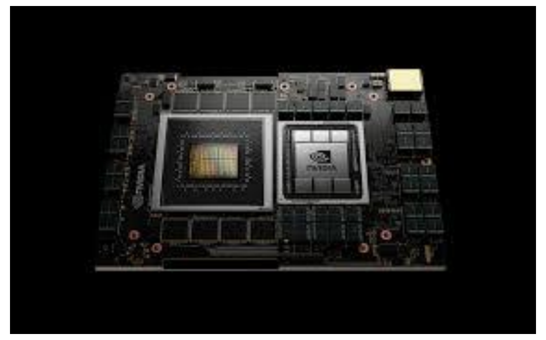

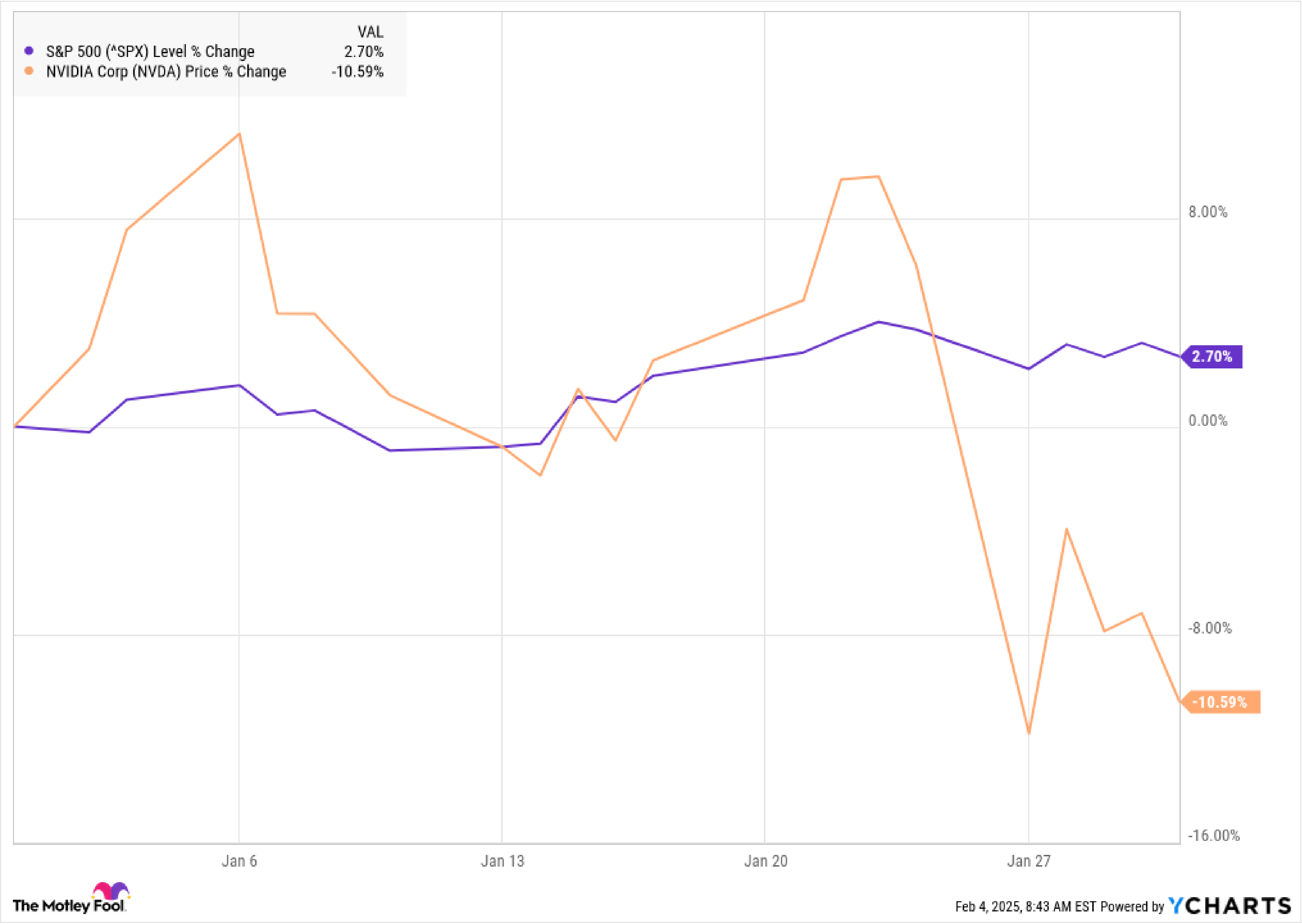

Why?

- DeepSeek model shows AI can be done with less need for expensive hardware.

- Less ***future*** demand for more sophisticated NVIDIA chips than previously-expected

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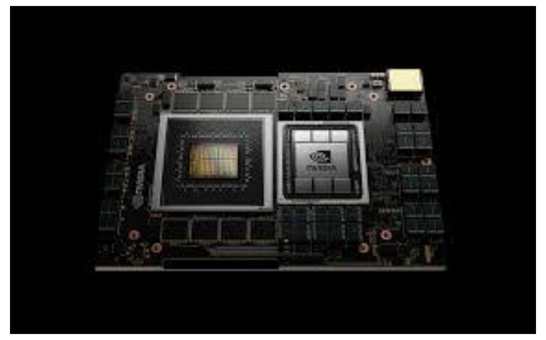

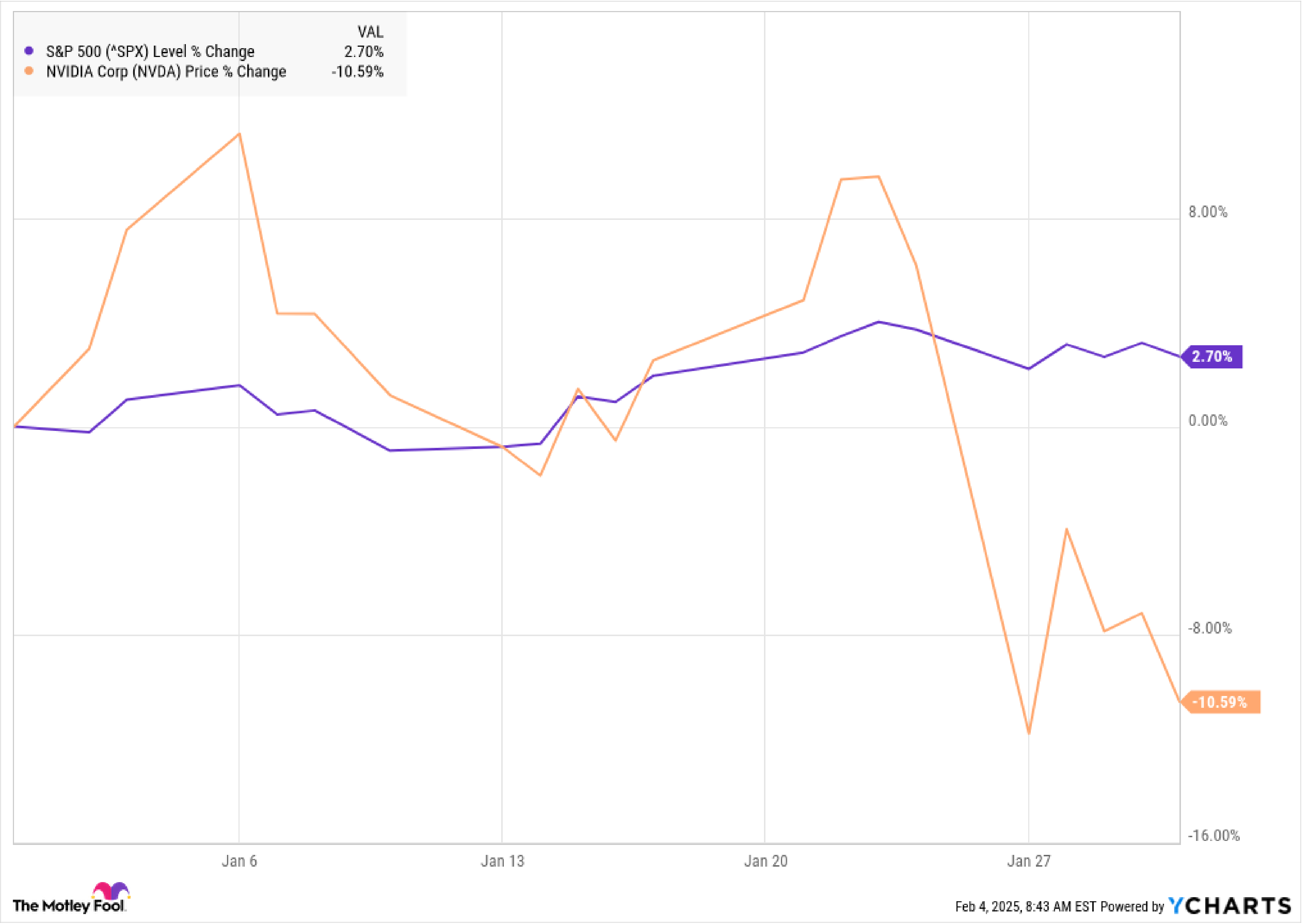

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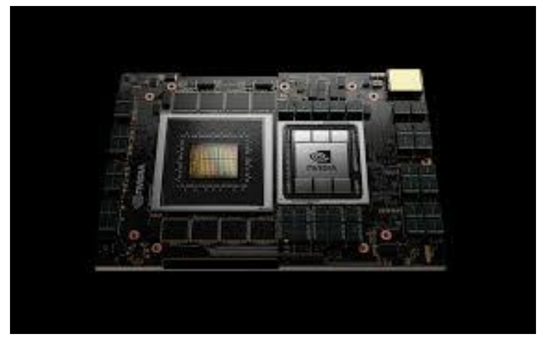

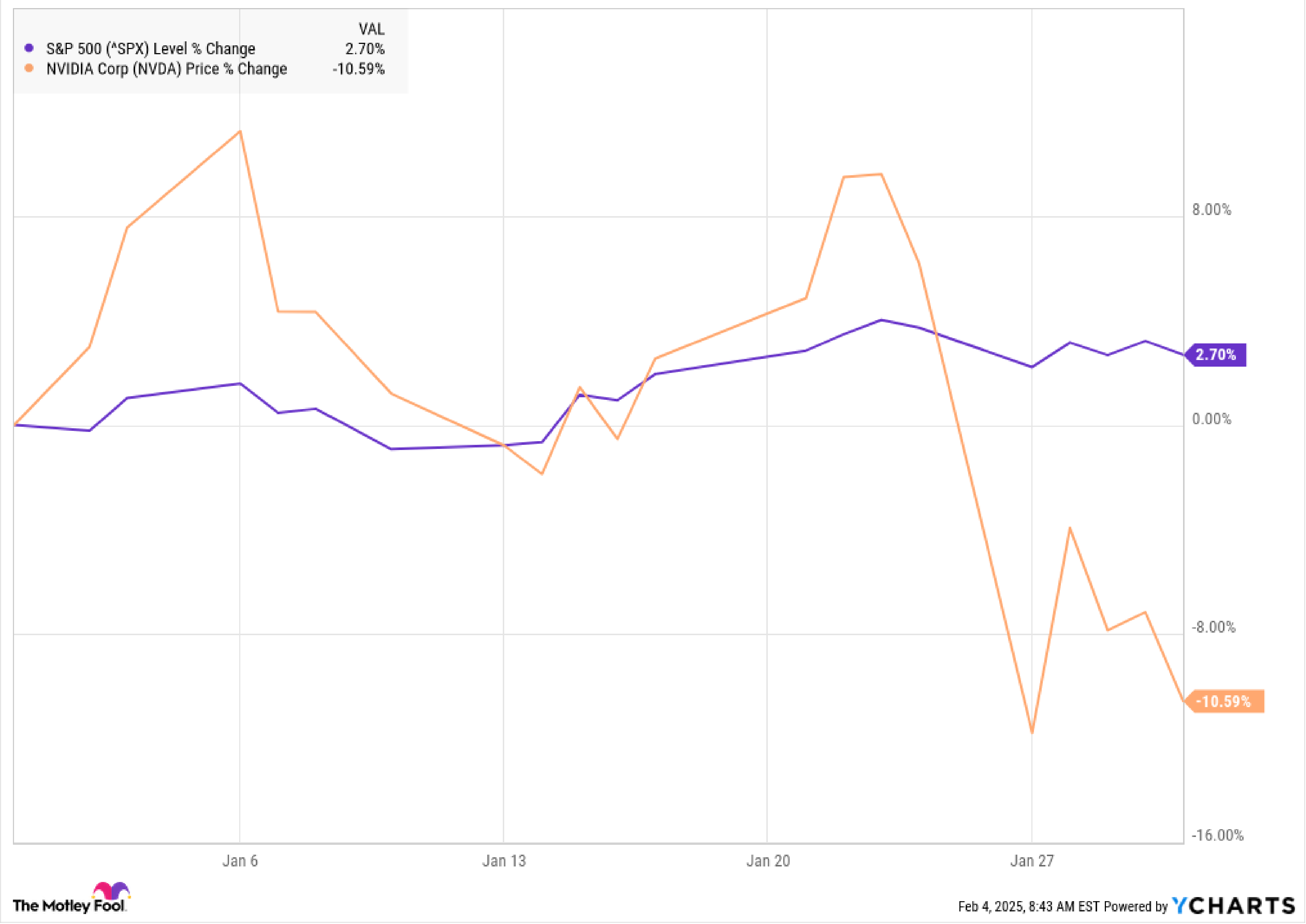

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Last week. Stock price falling despite 114% revenue growth in 2024. **Why?**

- Stock prices are ***forward-looking***.
- Investors are adjusting down their expectations of firm's ***future*** profits.

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How Did NASDAQ Stock Perform Recently? S&P500

Last 6 months to 28 February 2025

Last 12 months to 28 February 2025

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How Did Stocks Perform Recently? S&P500

Monthly Prices 1999 - 2025

Monthly Candlesticks 1999 - 2025

Warren Buffet

"If you aren't willing to own a stock for ten years, don't even think about owning it for ten minutes"

"When we own portions of outstanding businesses with outstanding managements, our favorite holding period is______________________.

Price-Earnings (P/E) Ratios

Price-Earnings (P/E) Ratio

- Price-Earnings (P/E) Ratio is ratio of firm’s PVGO to the value of its assets in place

- EPS is often shortened to E.

Price-Earnings (P/E) Ratio

- Price-Earnings (P/E) Ratio is ratio of firm's PVGO to the value of its assets in place

$$P = \frac{EPS_1}{r} + PVGO$$

$$\frac{P}{EPS_1} = \frac{1}{r}\left(1 + \frac{PVGO}{\frac{EPS_1}{r}}\right)$$

- EPS is often shortened to E.
- If you knew the P/E ratio you could estimate value by multiplying estimated earnings by P/E multiple.
- **"Comps":** Valuing one firm based on P/E of similar firms.
- **Enterprise Value Ratio: (Market value equity + Market value debt - cash)/EBITDA**
- EBITDA is earnings before interest, taxes, depreciation, and amortization.

Price-Earnings (P/E) Ratio in Practice

- Are low P/E ratio stocks cheap?
- Are high P/E ratio stocks expensive?
- Why do banks have low P/E ratios?

Company	2005	2013	2015	2020	2024
A	55	26	32	34	30
B	33	25	15	9	9
C	20	12	36	36	40
D	16	15	13	26	27
E	11	10	17	25	10
F	10	16	22	9	13
Average Historical S&P	15	15	22	16	33

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Price-Earnings (P/E) Ratio in Practice

- Are low P/E ratio stocks cheap?
- Are high P/E ratio stocks expensive?
- Why do banks have low P/E ratios?

Company	2005	2013	2015	2020	2024
A. Alphabet	55	26	32	34	30
B. Ebay	33	25	15	9	9
C. Microsoft	20	12	36	36	40
D. WalMart	16	15	13	26	27
E. Exxon Mobil	11	10	17	25	10
F. Citigroup	10	16	22	9	13
Average Historical S&P	15	15	22	16	33

Shiller PE Ratio for S&P500

Taking Stock

First Quarter

Week	Topics	After Class Assignments
1 **(17, 19 February)**	Time Value of Money	Quiz 1
2 **(24, 26 February)**	Capital Budgeting Tools, Bonds	Quiz 2
3 **(3, 5 March)**	Equities	Quiz 3
4 **(10, 12 March)**	Statistics, Risk and Return, Portfolio Theory **Beta Management Case is released after week 4 class.** **- Complete in usual MBA groups (on canvas).** **- Due before Week 6 class.** **- Will discuss in week 6 class.**	Quiz 4
5 **(17, 19 March)**	Capital Asset Pricing Model (CAPM) **Feedback survey, with week 11 topic suggestions**	Quiz 5
6 **(24, 26 March)**	Beta Management (Case), Course Review	Midterm